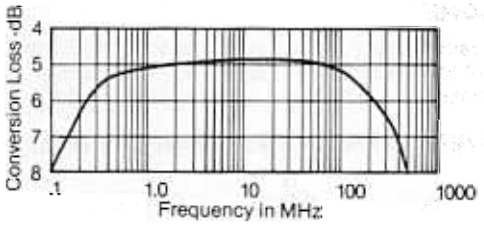
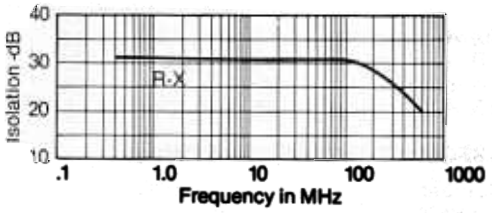
CM-1
Low Cost
Miniature
Double-
Balanced
Mixer
.5-500 MHz

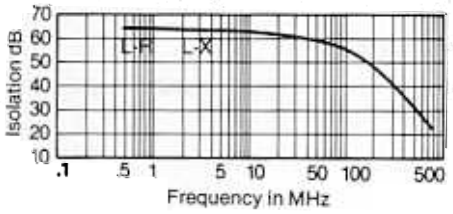
VARIL

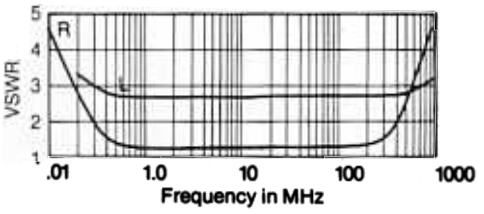
TYPICAL PERFORMANCE

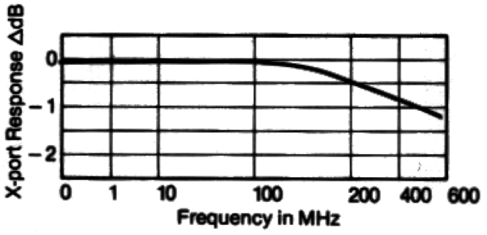
Impedance: All ports 50 ohms
1 dB Compression Point: 0 dBm
1 dB Desensitization Point: −2 dBm
3rd Order Intercept Point: +13 dBm
Noise Figure is within 1 dB of conversion loss
LO Power Range: +4 to +13 dBm











ENVIRONMENTAL CONDITIONS

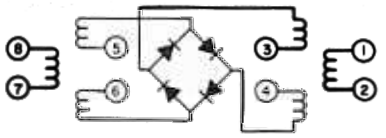
GUARANTEED ENVIRONMENTAL PERFORMANCE:

All units are designed to meet their specifications over −54°C to +100°C and after exposure to any or all of the following tests per MIL-STD-202E.

Exposure	Method	Test Condition
Thermal Shock	107D	B
Altitude	105C	G
H.F. Vibration	204C	D
Mechanical Shock	213B	C
Random Vibration (15 minutes per axis)	214	IIF
Solderability	208C	
Terminal Strength	211A	C
Resistance to Soldering Heat	210A	B

Sealed units, meet the requirements of Method 106D of MIL-STD-202E when exposed to humidity.

FUNCTIONAL SCHEMATIC



Pin Connections:

LO	8
RF	1
IF	3,4
Ground	2,5,6,7
Case Ground	2

NOTE: PINS 3 AND 4 MUST BE CONNECTED TOGETHER.
ALL GROUND PINS MUST BE GROUNDED.

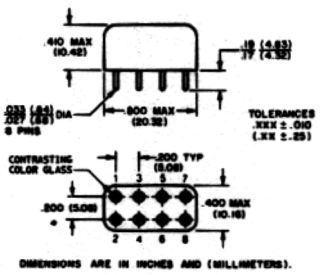
PACKAGE

MATERIAL:

Header: 1010 CRS
Pins: #52 Alloy
Seals: Glass
Cover: 18% Grade A Nickel Silver per ASTM B112-66, Alloy 2; QQ-C-585-1, Comp. 2, CDA-752 (65% Copper, 18% Nickel, 17% Zinc)

FINISH:

Cover: Nickel Silver
Header: Bright Tin Dip Per MIL-T-10727 Class II
Pins: Bright Tin Dip Per MIL-T-10727 Class II



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